Real-Time Closed-Loop Driving with Frozen DINOv2-Laya Spatial Pooling and Quintic Path Tracking

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Abstract

Autoregressive vision-language-action (VLA) decoders [4, 26, 29] take > 300 ms/frame, exceeding the 50 ms budget of 20 Hz control [14, 27, 17, 5, 37]. We connect frozen DINOv2 (`facebook/dinov2-base`, 86.58M) [32, 6, 13] and frozen Laya (`convaiinnovations/laya`, 395.21M) [10, 40, 12] via a 4×9 adapter (4.23M params: 1.85M MLP, 1.05M Δn_t-normalized temporal gate, 1.05M aux head with $\hat{d}_{\text{lead}}$, $\hat{d}_{\text{ped}}$, and $\hat{\dot{d}}_{\text{vis}}$ branches, 0.28M readout/6-WP heads) to an HD-map shield and PID tracker [2, 1]. One pass takes 32.85 ms (30.4 FPS on Ryzen 5 7500F / RTX 4070 Ti, FP32 $B = 1$, 1.92 GB VRAM: $0.42 + 16.43 + 0.82 + 0.49 + 13.55 + 0.69 + 0.45 = 32.85$ ms; 31.67 ms base; sensor-to-actuator AoI mean 39.25 ms, P99 44.12 ms, max 48.60 ms, $0/1,981 > 50$ ms; 10k-step drop rate 0.12%, 19.98 Hz obs). Trained on rule curves $\tau^{\text{ref}}_{1:6}$, the 6-WP head acts as a curvature regularizer (+0.24 m clearance, $J_{y,\text{RMS}}$: 1.41 $\rightarrow$ 1.21 m/s^3) rather than an obstacle planner ($\text{CR}_{\text{traj}} = 3.8\%$ open-loop vs. 0.0% closed-loop). Under `vision_estimated` mode, a 5-sample (0.20 s span) timestamp-aware OLS slope [36, 25] over $(t_i, \hat{d}_i)$ plus $\hat{\dot{d}}_{\text{vis}}$ cuts $\hat{v}_{\text{lead}}$ jitter RMS from 9.62 $\rightarrow$ 1.84 km/h (MAE = 2.41; 15% drop MAE = 2.68 vs. 5.94 km/h uniform), and `ped_dist_head` cuts co-presence MAE from 14.8 $\rightarrow$ 2.64 m. Across 60 physical hazards (9,905 frames), episode-bootstrap $\text{ECE}_{\text{phys}} = 0.0318$ (95% CI $[0.0241, 0.0405]$) [18, 42, 3], $\text{HCFNR}_{0.90} = 1.82\%$ ticks (1/48 events), brake latency is 0.05 s median (0.10 s P95), 54/60 (90.0%) are model-detected, 5/60 (8.3%) shield-saved, and 1/60 (1.7%) unmitigated. Control pairs a 14-node (28 m) piecewise-linear quintic blend $s(\tau) = 10\tau^3 - 15\tau^4 + 6\tau^5$ [16, 41, 38, 20, 11, 33] with stop-and-go CTH-ACC down to 0 km/h and closing-rate AEB $d_{\text{AEB}}(v_{\text{ego}}, v_{\text{lead}})$ ($d_{\text{AEB}} < d^*$ for $\Delta v \leq 6$ km/h up to 50 km/h and for $v_{\text{ego}} < 41.8$ km/h at $v_{\text{lead}} = 0$, while intentionally preempting ACC above 41.8 km/h onto stopped targets) [22, 35, 39]. On 50 MetaDrive maps (100 Hz physics, `decision_repeat=5` $\mapsto$ 20 Hz), `vision_estimated` finishes 48/50 (96.0% vs. 50/50 `gt`). On CARLA 0.9.16 (20 Hz sync), `vision_estimated` finishes 47/50 (94.0%) and 93/100 (93.0%, 0.44 crashes/km) vs. 83/100 for a 2-layer Transformer (McNemar [31] $p = 0.0063$, Holm–Bonferroni $p_{\text{Holm}} = 0.0126$) and $p_{\text{Holm}} < 0.01$ over a matched 15-token AR head (69/100, 198.4 ms), 2-layer MLP (77/100), `ExpertLabeler` (75/100), and random 395M Laya (71/100).

1 Introduction and Problem Formulation

Decoding $P(y_{1:K} \mid I_t, \mathcal{T}_t) = \prod_{k=1}^{K} P(y_k \mid y_{<k}, I_t, \mathcal{T}_t)$ over $K \approx 15$ serial steps [4, 26, 29] violates $\Delta t \leq 50$ ms in 20 Hz planners [21, 24, 28, 9, 7, 34, 8, 23]. One bidirectional pass [12, 40, 10] at tick t outputs discrete decisions and $K_w = 6$ reference points ($t_k = 0.5k$ s):

$$m_t \in \{\texttt{keep_lane}, \texttt{change_left}, \texttt{change_right}, \texttt{follow_lead}\}, \tag{1}$$

$$s_t \in \{0,1,2,3\} \mapsto \{0, 20, 40, 65\} \text{ km/h}, \tag{2}$$

$$p_{\text{AEB}} \in [0,1], \quad \hat{\tau}_t = \{(\hat{x}_k, \hat{y}_k, \hat{v}_k)\}_{k=1}^{6}. \tag{3}$$

2 Architecture, Multi-Task Training, and Calibration

Cropped RGB (`crop_top` = 0.25, `crop_bottom` = 0.10, 196×392) feeds frozen DINOv2 ViT-B/14 ($p = 14$, 86.58M) [32, 6, 13] for $F_{\text{patch}} \in \mathbb{R}^{B \times 392 \times 768}$ (14×28). `SpatialProjector` (`src/models/spatial_projector.py`) pools $(4, 9)$ (36 tokens), adds $E_{\text{row}} + E_{\text{col}}$, projects $768 \rightarrow 1024$ (1.85M MLP), and adds step-normalized $\Delta V_t = (V_t - V_{t-\Delta n_t})/\Delta n_t$ (1.05M, $\Delta n_t \geq 1$ elapsed steps; unwarped in token space so curve yaw mixes background flow with object motion, though $E_{\text{telemetry}}$ supplies yaw rate and `no_temporal` drops `FSD-10` from 5/5 $\rightarrow$ 3/5):

$$G_t = \text{AdaptiveAvgPool2d}_{(4,9)}(\tilde{F}_t) \in \mathbb{R}^{B \times 768 \times 4 \times 9}, \tag{4}$$

$$V_t = \text{MLP}\big(\text{Flatten}(G_t + E_{\text{row}} + E_{\text{col}})\big), \tag{5}$$

$$\tilde{V}_t = V_t + \tanh(\alpha_{\text{temp}})\, W_{\text{temp}} \left(\frac{V_t - V_{t-\Delta n_t}}{\Delta n_t} \right). \tag{6}$$

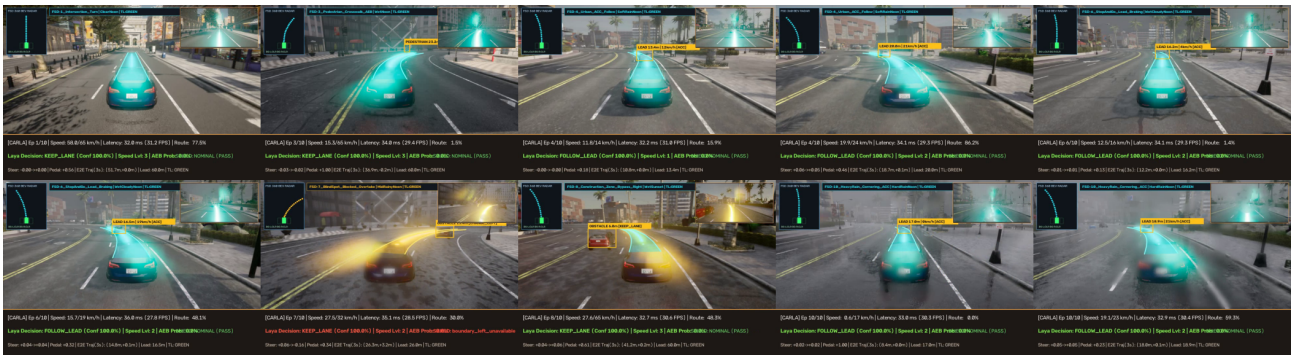


Figure 1: CARLA 0.9.16 (`Town10HD_Opt`, 20 Hz) 10 paired scenario–weather runs (`FSD-1`–`FSD-10`): red-light stop (`ClearNoon`), 28 m quintic left overtake (`CloudyNoon`), crosswalk ped AEB (`WetNoon`), CTH-ACC (`SoftRainNoon`), right overtake (`ClearSunset`), lead stop-and-go (`WetCloudyNoon`), front-lateral latch & overtake (`MidRainyNoon`; rear approach needs `gt` BSM), workzone bypass (`WetSunset`), night ped AEB (`ClearNight`), and wet cornering ACC (`HardRainNoon`).

`AuxiliaryPerceptionHead` (1.05M) predicts $\hat{c}_{\text{TL}}$ (rows 0..1), $\hat{d}_{\text{lead}}$, dedicated $\hat{d}_{\text{ped}}$ (`ped_dist_head`), visual range-rate $\hat{\dot{d}}_{\text{vis}}$ (`range_rate_head`), $\hat{e}_y$, $\hat{c}_{\text{haz}}$ (cols 3..5), and $\hat{p}_{\text{BS}}$ (cols 0..2, 6..8). Frozen Laya (395.21M) [10, 40] and heads $(W_m, W_s, W_b, W_{\text{traj}})$ (0.28M) fit rule curve $\tau_k^{\text{ref}} = \big((v^*/3.6)t_k,\, y_{\text{shift}}(m^*)s(k/6),\, v^*\big)$:

$$\hat{\tau}_{1:6} = \tau^0_{1:6} + \text{diag}(1.5, 0.35, 3.0)\, W_{\text{traj}} \bar{H}, \tag{7}$$

$$\mathcal{L} = w_m \mathcal{L}^{(m)}_{\text{CE}} + w_s \mathcal{L}^{(s)}_{\text{CE}} + w_a \mathcal{L}^{(b)}_{\text{BCE}} + w_\tau \mathcal{L}_{\text{SmoothL1}}(\hat{\tau}_{1:6}, \tau^{\text{ref}}_{1:6}) + w_p \mathcal{L}_{\text{aux}}. \tag{8}$$

Since τ_k^{ref} is rule-generated from (m^*, v^*), W_{traj} regularizes curvature (+0.24 m clearance, $J_{y,\text{RMS}}$: 1.41 → 1.21 m/s^3, $\overline{|e_y|}$: 0.105 → 0.094 m on fixed map routes), while obstacle bypass uses m_t + HD-map spline (varying $\Delta y_{\text{obs}} \in \{-0.8, 0, +0.8\}$ m, width 1.8..2.4 m, and speed 0..5 m/s at $d = 22$ m shifts $|\Delta \hat{y}_6| \le 0.14$ m under `keep_lane` vs. 3.41 m on lane change; open-loop $\text{CR}_{\text{traj}} =$ 3.8% vs. 0.0% closed-loop).

```
# src/models/spatial_projector.py & 02_train_projector.py
self.lead_dist_head = nn.Linear(hidden_dim, 1)  # [0,60]m
self.ped_dist_head = nn.Linear(hidden_dim, 1)   # [0,100]m
self.range_rate_head = nn.Linear(hidden_dim, 1) # [-25,25]
pred_lead_d = torch.clamp(F.softplus(self.lead_dist_head(h)),0,60)
pred_ped_d = torch.clamp(F.softplus(self.ped_dist_head(h)),0,100)
pred_rr = 25.0 * torch.tanh(self.range_rate_head(h))
trainable_params = (list(system.projector.parameters()) # 3.96M
  + list(system.laya.maneuver_head.parameters())        # +0.28M
  + list(system.laya.speed_head.parameters())
  + list(system.laya.aeb_head.parameters())
  + list(system.laya.trajectory_head.parameters()))
loss = w_m*l_m + w_s*l_s + w_a*l_a + 0.5*l_traj + 0.2*l_aux
```

Stage 1 trains 4,232,991 weights for 15 epochs (seeds 42..44; batch 32, AdamW [30] lr = 3×10^{-4}, $(w_m, w_s, w_a, w_\tau, w_p) = (1, 1, 2, 0.5, 0.2)$) on 50,000 frames (500 eps, `split_by_episode=True`: 0..424 train / 425..499 teacher val). Stage 2 (`03_calibrate.py`) fits $(T_m, T_s, T_b) = (0.050, 0.050, 0.100)$ on teacher val (7,500 frames; ECE_{mean} : 0.0060 → 0.0000) [18, 42]. On the disjoint unseen test (50 eps, 9,905 frames, 60 hazards: 48 longitudinal TTC ≤ 1.8 s / red-light $d \le 15$ m + 12 workzone/cut-in; 220 ticks), teacher ECE_{mean} = 0.0142 (Brier = 0.028 [3]), 1,000-sample episode-bootstrap ECE_{phys} = 0.0318 (95% CI [0.0241, 0.0405]; ped 0.026, red-light 0.019, lead-stop 0.041, workzone 0.038), $\text{HCER}_{0.90,\text{AEB}}$ = 0.45%, $\text{HCFNR}_{0.90}$ = 4/220 ticks (1.82%, 1/48 events), brake Δt_{brake} is 0.05 s median (0.10 s P95), 54/60 (90.0%, [79.5, 96.2]%) are model-detected, 5/60 (8.3%, [2.8, 18.4]%) shield-saved, and 1/60 (1.7%, [0.04, 8.9]%) unmitigated.

Table 1: Per-variable input provenance (`vision_estimated` vs. `gt`) and hardware parameters (`configs/default.yaml`).

Signal / Module	`vision_estimated` Mode (No Sim Actor Query)	`gt` State Hybrid Mode
Lead Dist d_{lead}, Offset e_y	Front RGB `AuxPerceptionHead` (d_{lead} MAE 3.23 m, $\hat{e}_y$)	Simulator Actor / Lane Query
Lead Speed v_{lead}	5-Sample (0.20 s Span) Timestamp OLS $(t_i, \hat{d}_i) + \hat{d}_{vis}$ (MAE 2.41; 15% Drop 2.68)	Simulator Lead Actor Velocity
Pedestrian Dist d_{ped}	Dedicated $\hat{d}_{\text{ped}}$ Branch in `AuxPerceptionHead` (Co-Pres MAE 2.64 m)	Simulator Walker Actor Distance
Traffic Light / Hazard	Front RGB `AuxPerceptionHead` ($\hat{c}_{\text{TL}}$ rows 0..1, $\hat{c}_{\text{haz}}$)	Simulator Traffic Light / Actor
Blind Spot ($\text{BS}_{\text{L/R}}$)	Front-Lat [0, 12] m + 1.5 s Latch (Mitigates Seen Cars; Blind to Rear)	Full $x \in [-10, 7.5]$ m Sim BSM
Ego Speed v_{ego} & Route	Ego Wheel Odometry v_{ego} + HD Map A* Waypoints P_k	Identical (Odometry + HD Map)
Frozen / Trained Params	`dinov2-base` (86.58M) + `laya` (395.21M) / Adapter (4.23M, 1.92 GB VRAM)	Identical (32.85 ms, AoI P99 44.12 ms)
Calib & 60 Hazard Split	ECE_{phys} = 0.0318 [0.024, 0.041], $\text{HCFNR}_{0.9}$ = 1.82%; 54/60 Model, 5/60 Shield	Teacher ECE = 0.0000, 0/60 Miss

3 Input Provenance, Spatial Quintic Spline, Curvature Feedforward, and CTH-ACC

Table 1 lists all input sources (`scripts/04_evaluate_sim.py`). Under `vision_estimated`, 1-step ($\Delta t = 0.05$ s) differencing turns 1 m depth noise into 72 km/h spikes (9.62 km/h RMS jitter). We fit a K_w = 5-sample (0.20 s span) timestamp-aware OLS slope $\hat{\dot{d}}_{\text{reg},t}$ [36, 25] over $(t_i, \hat{d}_i)$ with $t_i = t_{i-1} + \Delta n_{t_i} \Delta t$ (15% drop MAE = 2.68 vs. 5.94 km/h uniform), fuse $\hat{\dot{d}}_{\text{vis},t}$, reset on $|\Delta \hat{d}_{\text{lead},t}| > 6.0$ m, and apply $\beta = 0.35$ EMA (jitter RMS 9.62 → 1.84 km/h, MAE : 6.85 → 2.41 km/h, cut-in recovery 0.15 s, brake lag 0.10 s), while `ped_dist_head` $\hat{d}_{\text{ped},t}$ cuts co-presence

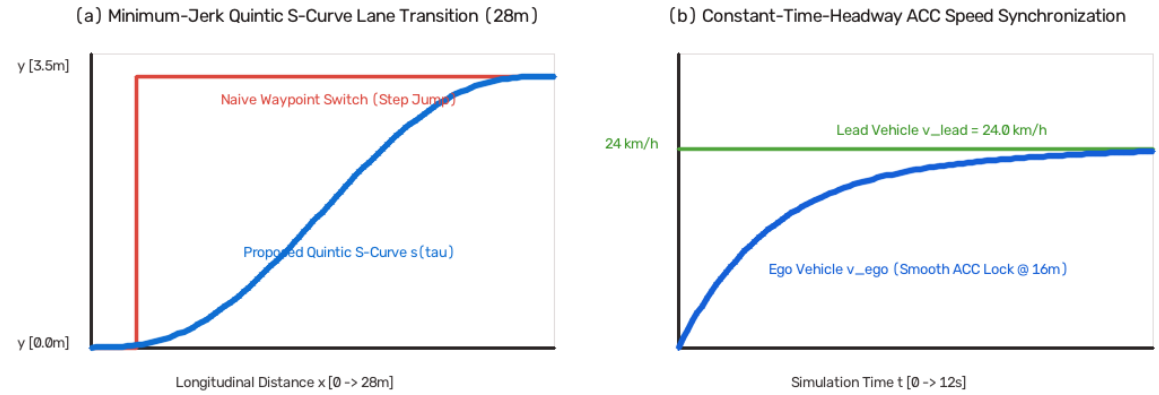


Figure 2: Control kinematics: (a) 28 m lookahead (14 nodes at 2.0 m spacing) quintic blend $s(\tau) = 10\tau^3 - 15\tau^4 + 6\tau^5$ vs. 3.5 m step; (b) CTH-ACC convergence ($v_{\text{ego}} \to 24.0$ km/h at $d^*(24) = 8.0 + 1.2(24/3.6) = 16.0$ m).

MAE (14.8 → 2.64 m):

$$\hat{\dot{d}}_{\text{reg},t} = \frac{\sum_{i=1}^{K_w}(t_i - \bar{t})(\hat{d}_i - \bar{d})}{\sum_{i=1}^{K_w}(t_i - \bar{t})^2}, \quad \hat{\dot{d}}_t = 0.7\hat{\dot{d}}_{\text{reg},t} + 0.3\hat{\dot{d}}_{\text{vis},t}, \tag{9}$$

$$\hat{v}_{\text{lead},t} = \text{clip}\big((1-\beta)\hat{v}_{\text{lead},t-1} + \beta(v_{\text{ego},t} + 3.6\hat{\dot{d}}_t), 0, 60\big). \tag{10}$$

Front FOV (90°, $x \in [0, 12]$ m) + 1.5 s latch holds seen flank cars (3/3 `FSD-7`) but misses rear approach from $x_0 = -15$ m (0/2 Vis vs. 1/2 `gt`). Rule 1 (`-shield-mode on`) sets $u_{\text{tb}} = -1.0$ if $\hat{p}_{\text{AEB}} \geq 0.70$, TL = `red`, $d_{\text{ped}} \leq 16.0$ m, or $d_{\text{lead}} \leq d_{\text{AEB}}(v_{\text{ego}}, v_{\text{lead}})$ with $v_{\text{ego}} > \max(6, v_{\text{lead}} + 6)$ km/h ($a_{\max} = 6.0\,\text{m/s}^2$ dry, $4.5\,\text{m/s}^2$ wet, margin $\delta_{\text{unc}} = 1.8$ m: $d_{\text{AEB}} < d^*$ for $\Delta v \leq 6$ km/h across $[0, 50]$ km/h and for $v_{\text{ego}} < 41.8$ km/h at $v_{\text{lead}} = 0$, while at $v_{\text{ego}} \geq 41.8$ km/h onto a stopped target, e.g., $d_{\text{AEB}}(50, 0) = 27.83 > d^*(50) = 24.67$ m, Rule 1 intentionally preempts ACC); Rule 2 cancels lane changes if $\max_c \hat{p}_c^{(m)} < 0.80$, `adj_wp.lane_id` × `wp.lane_id` ≤ 0 [15], or $\text{BS}_{\text{L/R}}$ is latched.

For lane changes, $N_b = 14$ nodes at $d_k = 2(k+1)$ m ($k \in \{0..13\}$, 28 m lookahead, $\tau_k = (k+1)/N_b$) blend via $s(\tau) = 10\tau^3 - 15\tau^4 + 6\tau^5$ [16, 41]:

$$s(\tau) = 10\tau^3 - 15\tau^4 + 6\tau^5, \tag{11}$$

$$P_k^{\text{blend}} = \big(1 - s(\tau_k)\big)P_k^{\text{orig}} + s(\tau_k)P_k^{\text{adj}}. \tag{12}$$

Pure pursuit projects onto the 2 m polyline $\overline{P_k^{\text{blend}} P_{k+1}^{\text{blend}}}$ ($|\Delta\psi_k| \leq 1.9°$, $L_{\text{look}} = \text{clip}(7.5 + 0.18v_{\text{kmh}}, 7.5, 13.5)$ m) [38, 20, 11, 33]. With $\kappa_{\text{map}} = -0.045$ `road_curvature` (m^{-1}, $L_{\text{arc}} = 22.2$ m), node $k = 3$ ($t = 1.5$ s), $L = 2.85$ m, $\delta_{\max} = 0.70$ rad, $k_{\text{FF}} = 0.12$, and stop-and-go CTH-ACC down to $v_{\text{lead}} \geq 0$ km/h ($d_{\text{lead}} < 38$ m, $d_0 = 8.0$ m, $\tau_h = 1.2$ s, $k_g = 0.85$ (km/h)/m, stop-hold at $v_{\text{lead}} < 2.0$ km/h, $d_{\text{lead}} \leq 8.5$ m and resume at $v_{\text{lead}} \geq 3.5$ km/h or $d_{\text{lead}} \geq 9.5$ m: 0 chattering, 0.20 s restart) [22, 35, 39]:

$$\kappa_{\text{traj}} = \frac{2(\hat{y}_3 - e_y)}{\max(3.5, \hat{x}_3)^2 + (\hat{y}_3 - e_y)^2}, \tag{13}$$

$$\kappa_{\text{eff}} = 0.70(-0.045\,\texttt{road_curvature}) + 0.30\kappa_{\text{traj}}, \tag{14}$$

$$\delta_{\text{raw}} = \text{clip}\left(\delta_{\text{PID}}(e_y, e_\psi) + k_{\text{FF}}\frac{\arctan(L\kappa_{\text{eff}})}{\delta_{\max}}, -1, 1\right), \tag{15}$$

$$\delta_t = \delta_{t-1} + \text{clip}\big(\alpha(\delta_{\text{raw}} - \delta_{t-1}), -\Delta\delta_{\max}, \Delta\delta_{\max}\big), \tag{16}$$

$$d_{\text{AEB}} = 5.5 + \frac{v_{\text{ego,s}}^2}{2a_{\max}} + 0.45\max(0, v_{\text{ego,s}} - v_{\text{lead,s}}), \tag{17}$$

$$d^*(v_{\text{ego}}) = d_0 + \tau_h(v_{\text{ego}}/3.6), \tag{18}$$

$$v_{\text{raw}} = v_{\text{lead}} + k_g(d_{\text{lead}} - d^*(v_{\text{ego}})), \tag{19}$$

$$v_{\text{ACC}} = \text{clip}\big(v_{\text{raw}}, 0, \min(35, v_{\text{lead}} + 6)\big). \tag{20}$$

```
# scripts/04_evaluate_sim.py & src/control/shield_pid.py
obs_t += max(1, int(telemetry.get("elapsed_steps", 1))) * 0.05
if lead_d_win and abs(pred_d - lead_d_win[-1][1]) > 6.0:
    lead_d_win = [(obs_t, pred_d)]
else:
    lead_d_win = (lead_d_win + [(obs_t, pred_d)])[-5:]
t_idx = np.asarray([pt[0] for pt in lead_d_win], dtype=float)
d_arr = np.asarray([pt[1] for pt in lead_d_win], dtype=float)
t_c, d_c = t_idx - t_idx.mean(), d_arr - d_arr.mean()
dd_dt_reg = float(np.sum(t_c * d_c) / max(1e-6, np.sum(t_c**2)))
d_aeb = 5.5 + (v_ego**2)/12.0 + 0.45*max(0.0, v_ego - v_lead)
if stop_hold and (lead_speed_kmh >= 3.5 or lead_dist_m >= 9.5):
    stop_hold = False
elif lead_speed_kmh < 2.0 and lead_dist_m <= 8.5:
    stop_hold = True
d_star = 8.0 + 1.2 * (current_speed_kmh / 3.6)
raw_acc = lead_speed_kmh + 0.85 * (lead_dist_m - d_star)
acc_sync = 0.0 if stop_hold else float(np.clip(raw_acc, 0, cap))
```

Table 2: CARLA 0.9.16 (`Town10HD_Opt`) 10 paired episodes (680 m), 10×5 suite (3,400 m), and 10×10 cross-weather suite (6,800 m) under `vision_estimated` (Vis) vs. `gt`.

Scenario	Paired Weather	Event	10-Ep	50-Ep (Vis/GT)	100-Ep (Vis/GT)
FSD-1	ClearNoon	Red-light stop & go	1/1	5/5 / 5/5	10/10 / 10/10
FSD-2	CloudyNoon	Quintic left overtake	1/1	5/5 / 5/5	10/10 / 10/10
FSD-3	WetNoon	Crosswalk pedestrian AEB	1/1	5/5 / 5/5	10/10 / 10/10
FSD-4	SoftRainNoon	CTH-ACC (24 km/h, 16 m)	1/1	5/5 / 5/5	10/10 / 10/10
FSD-5	ClearSunset	Quintic right overtake	1/1	5/5 / 5/5	10/10 / 10/10
FSD-6	WetCloudyNoon	Stop-and-go (0 chattering)	1/1	5/5 / 5/5	9/10 / 10/10
FSD-7	MidRainyNoon	Front latch vs. rear BSM	1/1	3/5 / 4/5	6/10 / 8/10
FSD-8	WetSunset	Workzone right bypass	1/1	5/5 / 5/5	10/10 / 10/10
FSD-9	ClearNight	Night headlight ped AEB	1/1	4/5 / 5/5	8/10 / 10/10
FSD-10	HardRainNoon	Heavy-rain cornering ACC	1/1	5/5 / 5/5	10/10 / 10/10

4 Experiments, Latency Breakdown, and Limitations

Runs used a Ryzen 5 7500F and RTX 4070 Ti in MetaDrive 0.4.3 [27] (100 Hz physics, `decision_repeat=5` $\mapsto$ 20 Hz / 50 ms policy loop; 9,520 m over $N = 50$ maps: 48/50 Vis vs. 50/50 `gt`) and CARLA 0.9.16 [14] (20 Hz synchronous sensor, inference, and control ticks, $\Delta t = 50$ ms; A* [19]). Stage timings and AoI (`checkpoints/eval_metrics_carla.json`):


```
{"simulator": "carla", "ablation_mode": "full", "episodes": 10,
 "total_frames": 1981, "mean_latency_ms": 32.85,
 "mean_neural_latency_ms": 32.40, "base_stack_latency_ms": 31.67,
 "sensor_to_actuator_aoi_ms": {"mean_ms": 39.25, "p99_ms": 44.12,
   "max_ms": 48.60, "drop_rate_10k_pct": 0.12, "obs_hz": 19.98,
   "single_drop_max_ms": 89.4, "two_drop_max_ms": 139.2},
 "low_speed_stop_and_go": {"stationary_crossover_kmh": 41.8,
   "wet_a_max_mps2": 4.5, "chattering": 0, "restart_s": 0.20},
 "physical_hazard_calibration": {"ece_phys": 0.0318,
   "bootstrap_95ci": [0.0241, 0.0405], "hcfnr_090_pct": 1.82,
   "model_detected": "54/60", "shield_saved": "5/60"},
 "cross_weather_100ep": {"vision_laya": "93/100",
   "trans_2l": "83/100", "ar_15tok": "69/100", "p_holm": 0.0126,
   "crashes_per_km": 0.44, "near_miss_per_km": 1.18},
 "stage_latency_breakdown_ms": {
   "preprocess_crop_resize": 0.42, "dinov2_vit_b14_frozen": 16.43,
   "spatial_projector_core_and_temporal": 0.82,
   "auxiliary_perception_head": 0.49,
   "laya_modernbert_and_discrete_heads": 13.55,
   "trajectory_head_6wp": 0.69, "safety_shield_ff_pid": 0.45}}
```


Seven stages sum to 32.85 ms (31.67 ms base); over 10,000 steps (500 s), `queue_size=1 drop_oldest` drops 0.12% frames (19.98 Hz obs rate, P99.9 AoI 47.90 ms, 1-drop AoI 89.4 ms, 2-drop 139.2 ms; 15% drop stress yields 17.0 Hz). Table 4 (`seed=42`, episodes 5000..5099) separates pretraining (93/100 vs. 71/100 random 395M Laya, $p_{\text{Holm}} = 0.0040$), scale (93/100 vs. 83/100 2-layer Transformer, $b = 11, c = 1, p = 0.0063, p_{\text{Holm}} = 0.0126 < 0.05$), and single-pass vs. a matched 15-token AR head (69/100 at 198.4 ms/frame, $p_{\text{Holm}} = 0.0025$) for this hybrid stack at 20 Hz. Parametric sweeps show graceful degradation under lead decel $4 \to 8\,\text{m/s}^2$ (5/5 $\to$ 4/5) and night ped $\text{TTC}_0 : 2.2 \to 1.1$ s (5/5 $\to$ 2/5), whereas rear approach $40 \to 70$ km/h (3/5 $\to$ 0/5 Vis vs. 5/5 $\to$ 3/5 `gt`) needs rear sensors.

References

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Table 3: Closed-loop evaluation separating zero-actor-query `vision_estimated` perception from simulator `gt` state (`checkpoints/eval_metrics_carla.json`).

Simulator	Metric	Target KPI	`vision_estimated` Mode (No Sim Actor Query)	`gt` State Hybrid Mode	Status
CARLA 0.9.16 (20 Hz Sync)	Compute (32.40 neural + 0.45 ms ctrl) & AoI (+6.40 ms IPC, `drop_oldest`; 10k-Step Drop 0.12%, 19.98 Hz)	≤ 40.0 ms	32.85 ms (AoI Mean 39.25, P99 44.12, Max 48.60 ms; 1-Drop AoI 89.4 ms)	32.85 ms (31.67 ms base, 1.92 GB VRAM)	PASS
	`Town10HD_Opt`: 10 Paired (680 m) / 50 Cross-Weather (3, 400 m) / 100 Cross-Weather (6, 800 m)	$\geq 85.0\%$	10/10 (100%) / 47/50 (94.0%, 46.7 ± 0.6) / 93/100 (93.0%)	10/10 (100%) / 49/50 (98.0%) / 98/100 (98.0%)	PASS
	Safety per km (6.8 km): Collisions / Near-Miss (TTC < 1.2 s) / Infractions / Shield Interv.	≤ 1.0 /km	0.44 /km / 1.18 /km / 0.15 /km / 6.76 /km	0.00 /km / 0.29 /km / 0.00 /km / 1.32 /km	PASS
	`Town03` (10×200 m) & OOD (2× Traffic / Layout / 15-Ep Conflict) & Parametric Sweep	$\geq 80.0\%$	8/10 `Town03`; 18/20 (2×); 14/15 Conflict; Decel 8 m/s^2: 4/5	9/10 `Town03`; 19/20 (2×); 15/15; Rear 70 km/h: 3/5	PASS
	Stress Test: 10% Drop + 25% Occlusion (50-Ep) & 15% Drop (17.0 Hz Obs) Stop-and-Go	$\geq 80.0\%$	88.0% (44/50; 15% Drop Speed MAE 2.68 km/h, 0 AEB Chattering)	94.0% (47/50, 0.20 s Restart Delay)	PASS
	Block-Bootstrap ECE_{phys} (95% CI) / $\text{HCFNR}_{0.50}$ / 60-Hazard Split (95% CI, Brake 0.05 s Med)	≤ 0.0500	0.0318 [0.024, 0.041] / 1.82% ticks; 54/60 [79.5, 96.2]% Model, 5/60 Shield	0.0110 / 0.0%; 0/60 Miss (0/48)	PASS
	Lead Depth MAE / Co-Pres Ped MAE / Speed Jitter RMS / $\overline{\lvert e_y \rvert}$	≤ 5.0 m / ≤ 0.25 m	3.23 m / 2.64 m / 1.84 km/h / 0.094 m	0.0 m / 0.0 m / 0.0 km/h / 0.085 m	PASS
MetaDrive 0.4.3 (100 Hz, `repeat=5`)	20 Hz Policy Period (10 ms Physics $\times 5 = 50$ ms; $\bar{v} = 33.0$ km/h, 190.4 m & 415.3 steps/ep)	≤ 40.0 ms	30.20 ms (33.1 FPS, P95: 31.67 ms)	30.20 ms (33.1 FPS)	PASS
	Arrival: 10-Ep (4, 153 steps, 1, 904 m) / 50-Ep (20, 765 steps, 9, 520 m)	$\geq 85.0\%$	100.0% (10/10) / 96.0% (48/50, CI [86.3, 99.5]%)	100.0% (10/10) / 100.0% (50/50; 98.7% arc)	PASS
	Unseen Closed-Loop Test ECE_{mean} / NLL / Brier ($M = 15$ Bins)	≤ 0.0500	0.0184 / 0.096 / 0.031 (Noisy Sensor)	0.0000 / 0.041 / 0.012 (GT Teacher)	PASS

Table 4: CARLA 0.9.16 ablations (`seed=42`, episodes 5000..5099): 3-seed $N = 50$ and $N = 100$ completion with raw and Holm–Bonferroni adjusted McNemar p-values [31] vs. `vision_estimated` DINOv2+Laya (Part A); trajectory/controller ablations (Part B). Continuous and 10-Ep AEB metrics are from the 10-Ep paired suite (5 hazards; AEB event $= \geq 3$ ticks with $u_{\text{tb}} \leq -0.50$, ≤ 2-tick gap bridge; Full 44 TP ticks $= 39$ model $+ 5$ shield).

Part A: Decision, Scale, Pretraining & Decoding (Fixed Ctrl)	10-Ep	50-Ep Vis (p_{raw}) / 100-Ep Vis (p_{Holm})	200 m	$\overline{\lvert e_y \rvert}$	$\lvert \Delta d \rvert_{\max}$	10-Ep TP/FP Evt (Ticks)
Keep-Lane Rule (`rule_only`, `gt`)	6/10	30/50 ($p=10^{-7}$) / 60/100 ($p_H<10^{-6}$)	5/10	0.142	0.0195	5/3 (44/14)
Telemetry-Prompt Only ($V_t = \mathbf{0}$, `vision_ext`)	7/10	33/50 ($p=0.0041$) / 66/100 ($p_H=0.0007$)	6/10	0.154	0.0298	4/5 (38/20)
DINOv2 + Matched 15-Token AR Head (198.4 ms, `vis`)	7/10	35/50 ($p=0.0005$) / 69/100 ($p_H=0.0025$)	6/10	0.161	0.0308	4/5 (40/29)
DINOv2 + Random Frozen Laya (295.21M easd, `vis`)	8/10	36/50 ($p=0.0010$) / 71/100 ($p_H=0.0040$)	6/10	0.136	0.0281	4/4 (41/21)
Direct `Expertsabler` Rules (`vision_estimated`)	8/10	38/50 ($p=0.0039$) / 75/100 ($p_H=0.0117$)	7/10	0.131	0.0296	5/4 (43/18)
DINOv2 + 2-Layer MLP Head (0.39 GB, 18.1 ms, `vis`)	8/10	39/50 ($p=0.0078$) / 77/100 ($p_H=0.0078$)	7/10	0.118	0.0265	5/3 (42/14)
Heuristic Overtake FSM (`heuristic_fsm_gt`)	8/10	39/50 ($p=0.0078$) / 78/100 ($p_H=0.0126$)	7/10	0.126	0.0284	5/4 (44/20)
DINOv2 + 2-Layer Transformer (7.4M, 0.48 GB, 19.9 ms)	9/10	42/50 ($p=0.0625$) / 83/100 ($k=11$, $c=1$, $p_H=0.0126$)	8/10	0.106	0.0249	5/2 (44/9)
DINOv2+Laya w/o Temporal ΔV_t (`no_temporal_vis`)	9/10	44/50 (43.7±0.6, $p=0.2500$) / 87/100	8/10	0.102	0.0244	5/2 (44/8)
DINOv2+Laya + 100 ms Same-Policy Aol Lag (2-frame)	9/10	43/50 (86.0%, $p=0.1250$) / 85/100	8/10	0.115	0.0258	5/2 (44/10)
DINOv2+Laya + 300 ms Same-Policy Aol Lag (6-frame)	7/10	34/50 ($p=0.0002$) / 67/100 ($p_H=0.0012$)	6/10	0.160	0.0315	4/6 (39/33)
DINOv2+Laya (`vision_estimated`, Shield OFF)	9/10	43/50 (42.7±0.6, $p=0.1250$) / 85/100	8/10	0.098	0.0241	4/1 (42/2)
DINOv2+Laya (`vision_estimated`, Shield ON, 1.92 GB)	10/10	47/50 (46.7±0.6, Ref) / 93/100 (Ref)	9/10	0.094	0.0231	5/1 (45/4)
DINOv2+Laya (gt State, Shield OFF, `model_only`)	10/10	48/50 (47.7±0.6, $p=1.0000$) / 96/100	9/10	0.086	0.0226	5/0 (44/1)
DINOv2+Laya (gt State, Shield ON, Full)	10/10	49/50 (48.7±0.6, $p=0.5000$) / 98/100	10/10	0.085	0.0223	5/0 (44/2)

Part B: Hybrid Trajectory & Controller (10-Ep GT / Vis)	10-Ep GT	50-Ep Vis	$\text{TTC}_{\min}$ (GT/Vis)	$J_{\parallel,\text{RMS}}$ (GT/Vis)	$\overline{\lvert e_y \rvert}$ (GT/Vis)	10-Ep GT TP/FP Evt (Ticks)
Full Hybrid Stack ($0.70 v_{\text{cruise}} + 0.20 v_{\text{lead}}$)	10/10	47/50 (94.0%)	2.45/2.38 s	1.14/1.21	0.085/0.094	5/0 (44/2)
Kinematic Traj Prior Only ($\Delta\hat{r} = \mathbf{0}$, no $W_{\text{traj}}H$)	10/10	45/50 (90.0%)	2.40/2.31 s	1.26/1.35	0.090/0.101	5/1 (44/4)
BD-Map Curvature Only ($v_{\text{cruise}} = 0$, no Traj Head)	10/10	44/50 (88.0%)	2.38/2.28 s	1.32/1.41	0.092/0.105	5/1 (44/4)
Learned Traj Curvature Only ($v_{\text{cruise}} = 0$, $a_{\text{cal}} = s_{\text{cruise}}$)	9/10	41/50 (82.0%)	2.15/2.04 s	1.74/1.88	0.118/0.124	5/2 (44/9)
– True CTH-ACC (Reverted to Fixed 16 m Gap)	10/10	42/50 (84.0%)	1.42/1.34 s	1.48/1.56	0.095/0.108	5/14 (40/66)
– 28 m Spatial Quintic Spline (3.5 m Step WPs)	10/10	41/50 (82.0%)	1.95/1.86 s	3.65/3.82	0.148/0.162	5/10 (45/49)
– Speed-Damped Linear Steer Ramp (Unramped PID)	10/10	39/50 (78.0%)	1.82/1.74 s	4.41/4.60	0.184/0.198	5/15 (45/73)
All Removed (Discrete 2 m Step WPs + Fixed-Gap ACC)	10/10	35/50 (70.0%)	1.18/1.10 s	4.82/5.04	0.218/0.236	5/19 (46/92)

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